Feasibility of Methods for Effective Detection and Classification of Sperm Whale Clicks in Large Passive Acoustic Monitoring Datasets

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Background

Sperm whales are regularly encountered in Canadian North Atlantic waters (Gomez et al., 2020), where they face risk of entanglement in Canada's North Atlantic fisheries (Feyrer et al., 2025), and other emerging threats, such as potential offshore wind development. Globally, this species is also still recovering from whaling (Whitehead and Shin, 2022) and is considered vulnerable by the International Union for Conservation of Nature (Taylor et al., 2019). However, the extent and seasonality of sperm whale presence in Canadian Atlantic waters is not well characterized.

Passive acoustic monitoring (PAM) is a powerful tool for monitoring spatial and temporal patterns in occurrence of cetacean species in general, and sperm whales in particular (e.g., Stanistreet et al., 2018), given that individuals regularly produce loud, broadband clicks. Fisheries and Oceans Canada (DFO) Maritimes Region collects extensive PAM data for monitoring of cetacean species that could be used to assess the spatial-temporal occurrence of sperm whales off Nova Scotia, yet sperm whale presence has been not routinely assessed in these datasets, largely due to a lack of a known efficient workflow for detecting and validating sperm whale clicks on these large PAM datasets.

To help to address this gap, this report evaluates possible approaches for the detection and classification of sperm whale clicks in large PAM datasets and provides recommendations on which approach(es) would be most suitable for efficient assessment of daily presence of sperm whale clicks on DFO's datasets.

Assessment Approach

Characterizing Dataset Parameters

I characterized the range of key parameters of relevant datasets (e.g., dataset size, recording schedules, sampling rate). I focused on acoustic datasets from Canadian waters (primarily from the North Atlantic Ocean) collected by DFO or by academic research groups whose data are available to me through research collaborations. My intention was not to create an exhaustive list of all datasets and parameters, but to capture a representative sample of the range of parameters in datasets of relevance for detection of sperm whales. When deciding which recordings were likely to be most relevant for sperm whale detection, I considered the properties of sperm whale echolocation clicks (which are the vocalization they most consistently produce). While sperm whales clicks have been detected in recordings with sampling rates as low as 1 kHz, with usable sound up to 470 Hz (Mellinger et al., 2004), typical centroid frequences of these clicks are 10-15 kHz (depending on recording aspect) and the –10 dB spectral endpoints are roughly 5 and 24 kHz (Madsen et al., 2002). Thus, I considered datasets with sampling frequencies below 12 kHz (i.e., bandwidths < 6 kHz) to be potentially less reliable for detecting sperm whale clicks and focused on recordings with sampling frequencies ≥ 12 kHz.

Identifying Factors for Evaluation

With input from the DFO Project Authority, I determined the primary factors of interest to evaluate potential efficiency and feasibility of different methods. The following factors were selected for assessment:

- useability (i.e., how stable, user friendly, well-documented)

- flexibility (i.e., ability to accommodate diversity of available datasets)
- computation time
- manual effort required
- performance metrics (e.g., false positive rate)
- software dependencies
- ease of access (i.e., monetary cost, readily available implementation)

Literature Review

I conducted a targeted literature review to identify studies that have assessed sperm whale clicks on large PAM datasets, or describe methods with the potential to do so. Potentially relevant publications were identified through multiple sources: a Web of Science (WoS) query (Feb 18, 2026; search query: Physeter macrocephalus acoustic*) and applicable references therein, from a list of methods reported in a recent review of echolocation click detection methods (Gracic et al., 2025), expert recommendation, and automatic Google Scholar alerts (for "Physeter macrocephalus").

Due to a large number of identified potentially relevant publications, I focused on recent literature (excluding publications from 2010 and earlier, unless the method described in that publication was used in one or more publications after this cut-off). Details from Tables 1-3 in the review by Gracic et al. (2025) were used to screen out methods that were not likely to be suitable (e.g., Cons mentioned high computational demand).

I then filtered search results for relevance based on title, abstract, and full publication, successively. To be considered relevant, the publication was required to: address detection and/or classification of sperm whale clicks using methods other than fully manual assessment of unprocessed data and not require a specific array set-up not currently used by DFO Maritimes. Studies describing methods focused on localization were also excluded, because they were expected to be more computationally demanding than necessary for detection and classification.

From these publications, I then focused on adequately described, available, and distinct methods. Thus, I excluded publications for which the method used was not described in sufficient detail and/or not clearly readily available (i.e., would require requesting code from authors, or developing an implementation from scratch based on the described methods), or which used a method described in another identified publication without meaningful variation. This screening process resulted in a list of sperm whale click detection-classification method options for detailed consideration.

Assessment of Methods

I assessed the shortlisted method options according to the factors of interest identified above. This assessment was based on the information gathered from the literature review, trials of the software and perspectives shared by other researchers with familiarity using these software systems. For trials of the software, I used sample datasets described in Tables A1 and A2 and software versions indicated in Table A3, run on a desktop computer (CPU: Processor AMD Ryzen 9 7950X3D 16-Core Processor, Memory: 96 GB 6000 MT/s DDR5, Disk: Samsung SSD 990 EVO Plus 1TB, GPU: AMD Radeon RX 9070 [16GB GDDR6 memory]). Further, I considered additional workflow steps (e.g., down sampling) that may be beneficial or necessary to pair with identified method options.

Outcomes

Literature Review

After removing duplicates (n = 13), I identified 307 potentially relevant documents, through Web of Science (n = 254), Gracic et al.' s 2025 review (n = 60), and other sources (n = 6), as shown in Figure A1. After filtering documents filtered based on recency (n = 72), information provided in Gracic et al. (2025) tables (n = 22), title (n = 53), abstract (n = 56), and full text (n = 34) and excluding two documents with no full text available, 68 documents remained. Further screening to exclude methods that were not described in sufficient detail (n = 9), were not clearly readily available (n = 21), used a method described in another identified publication without meaningful variation (n = 14), or used legacy software that has been superseded (n = 1) resulted in 23 shortlisted documents.

Among the shortlisted documents, several software platforms were used for detection of sperm whales (Table 1). Some documents used multiple software platforms, and multiple detection methods were used for some platforms. Most documents used some form of click detector (n = 13) and many used long-term spectral averages (LTSAs; n = 9). Methods using click detectors usually applied high pass, low pass or bandpass filters (typically from 1 or 2 kHz to 12 to 24 kHz) prior to analysis, while LTSA approaches did not. Detection and classification methods described in the shortlisted documents are examined in more detail in the Summary of Available Options, below.

Table 1. Software used for acoustic detection of sperm whales in shortlist of documents.

Software	Detection Method	Rate of Use (# of short-listed documents)[1]	Primary Reference
PAMGuard	Click detector LTSA	9 2	Gillespie et al., 2026
Triton	LTSA Click detector	5 1	Scripps Whale Acoustic Lab, Scripps Institution of Oceanography, La Jolla, CA
DetEdit	LTSA + Click detector	3	Solsona-Berga et al., 2020
CABLE	Click detector	1	Beslin et al., 2018
Raven Pro	Band-limited energy detection (BLED)	1	K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology, 2026
custom MATLAB script	Click detector	1	Gubnitky and Diamant, 2024
custom Python script	Neural Network	1	Bermant et al., 2019
unspecified	LTSA	1	N/A

[1] Due to the filtering that generated this shortlist, these relative rates of use may not be representative of frequency of use for analysis of sperm whale presence in acoustic data (e.g., several custom MATLAB script-based methods were filtered out due to not being clearly readily available).

Key Dataset Parameters

Datasets characterized (Table A1) included recordings from bottom-mounted moorings from the Northwest Atlantic (n = 5) and Northeast Pacific Oceans (n = 3) owned by DFO, a vessel-towed array operated by the Whitehead Lab at Dalhousie University in the Northwest Atlantic (n = 2), and a Slocum glider deployed in the Arctic Ocean for work led by the Fortune Whale Lab at Dalhousie University (n = 2). Some recorded continuously, while others were duty cycled, with deployment lengths of 9 to 384 days, and sampling frequencies of 16 to 256 kHz (Table A1).

Multi-day samples from four North Atlantic Ocean datasets were selected from among the available datasets to trial method processing speed and usability (Figure 1; Table A2). Two were duty cycled and two were continuous. Three were from single channel bottom-mounted moorings and one was from a two-element vessel-towed array. The sampling frequencies of these datasets ranged from 128-256 kHz, and some trials were run after decimating the files to lower sampling frequencies, as described below.

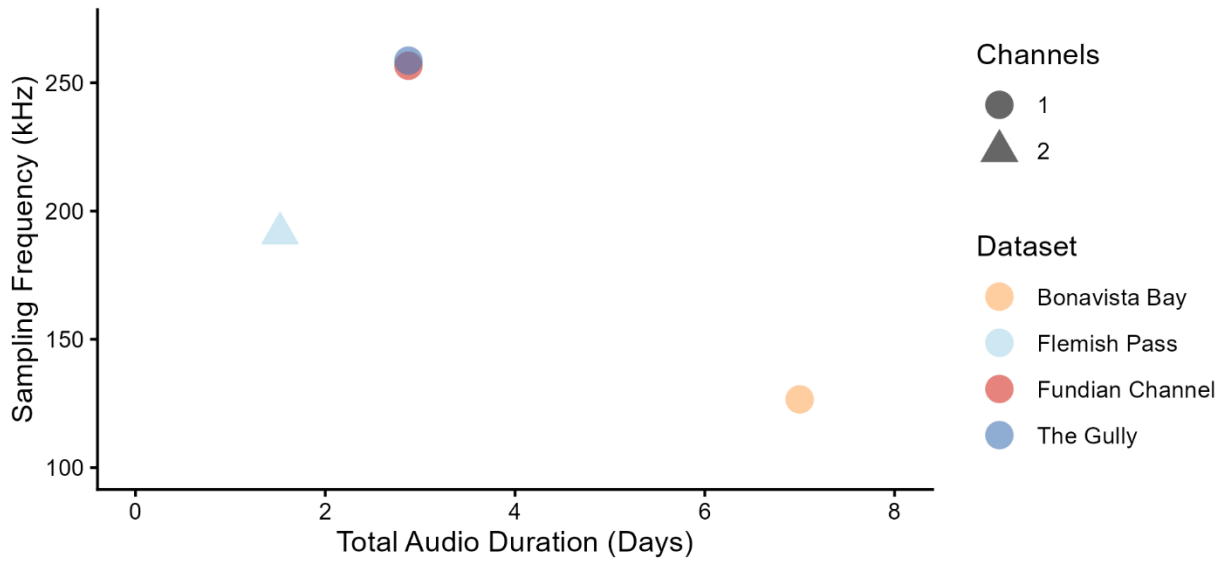

Figure 1. Overview of parameters of acoustic dataset samples from the Northwest Atlantic Ocean, used for method trials (see Table A2 for additional details). Point positions were slightly jittered, to allow overlapping points to be seen.

Summary of Available Options

Here, I focus on describing and assessing options of methods for detection and classification of sperm whale clicks from an operational perspective. Differences in technical approaches used for click detectors and click detection theory are described elsewhere (e.g., Gracic et al., 2025).

Some functionality was common across all methods considered. All methods were able to accommodate single channel recordings, though some (e.g., PAMGuard) are equipped to leverage multiple channels to localize individuals. All methods are applicable for stationary and mobile (e.g., towed or glider) recordings systems. All methods allow for batch processing of files.

Processing speed varied greatly across methods and, for some methods, across datasets (Figure 2). Raven's BLED was consistently the fastest at processing data, followed by Triton's LTSA or CABLE, depending on the dataset. CABLE also had the most variable processing speeds, ranging from 5 to 85 times faster than real time.

Further details on each method assessed are provided below.

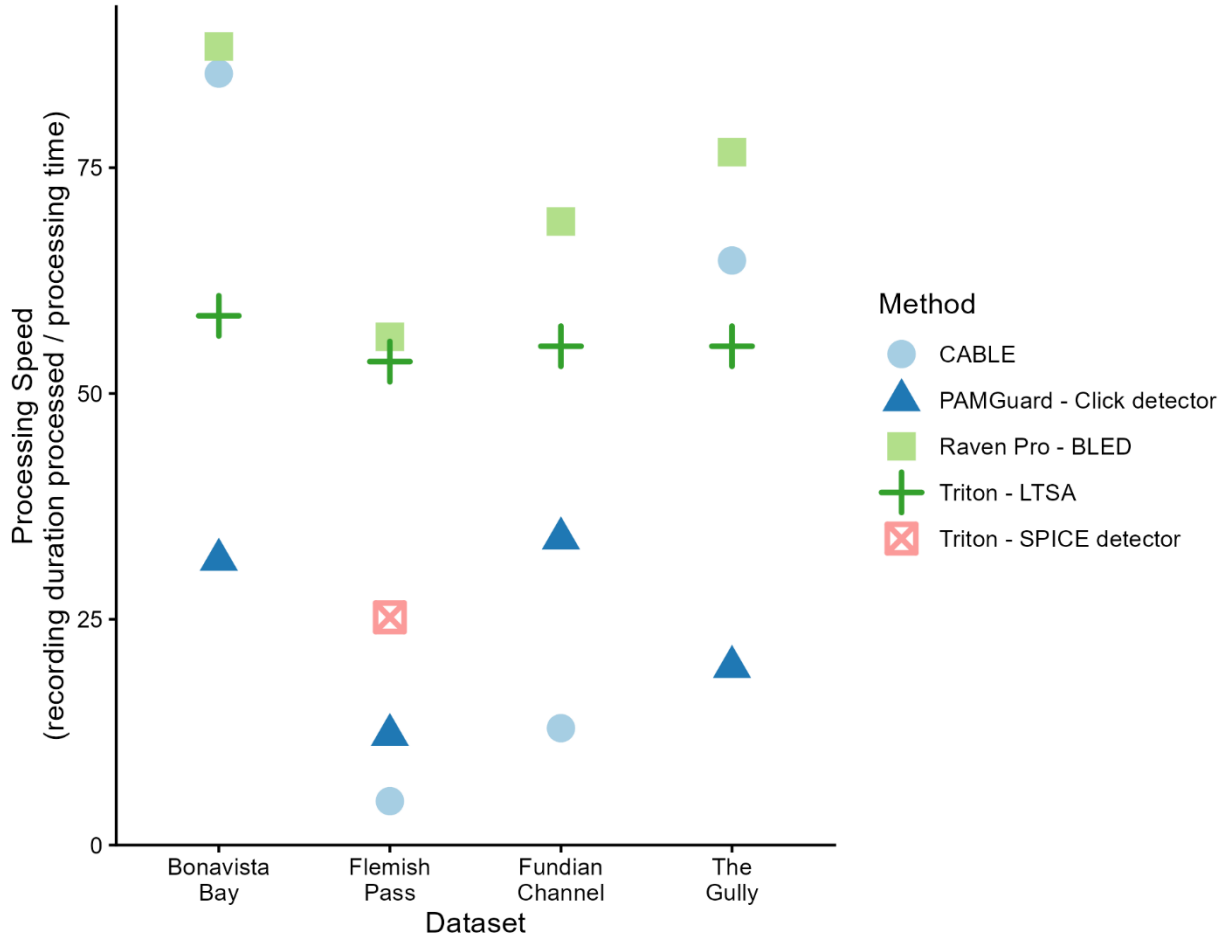

Figure 2. Processing speed across four datasets and five methods for detection of sperm whale clicks, run on a high-performance desktop computer. Raven Pro and Triton SPICE detector trials were completed using files that had already been decimated in Triton; for compatibility, speeds reported for these methods include the time spent on decimation. Only one dataset was successfully processed by Triton's SPICE detector. Speeds do not account for manual review of outputs.

PAMguard

PAMGuard is widely used for the acoustic detection of sperm whales (and other cetaceans), with features including click detectors, click train detectors, classifiers, and the ability to produce LTSAs. Classifier options include a multi-hypothesis tracking (MHT) click train classifier that groups clicks into trains based on inter-click interval (ICI), amplitude, frequency content and bearing.

Useability

Extensive documentation exists for PAMGuard (www.pamguard.org/olhelp/), but the learning curve is steep due to the complexity of the software, a complex user interface that is not always intuitive, and the need to set up configuration files that are aligned with your specific data inputs before being able to run the software. As an example of the user experience, I was able to follow the online documentation to run the generation of LTSAs fairly easily, but then could not figure out how to view them, despite following the documentation. This left me unsure of whether I was encountering a bug, or missing an undocumented step.

PAMGuard requires click length parameters of the click detector to be input in number of samples, rather than a unit of time, such as ms. While the conversion from ms to samples is straightforward for a known sampling interval, this adds some complexity when working with files of multiple different sampling frequencies. The most straightforward approach in such a case is likely to decimate all files to the same sampling frequency before further processing (as I did in my trials); an alternative would be to adjust these parameters for each set of files with a different sampling frequency.

As this software is still actively being updated, and new features added, bugs are not uncommon. However, bugs encountered in one version may be able to be resolved by switching to a different version of the software, and the software is still actively supported and bug reports are accepted.

Flexibility

PAMGuard is highly flexible to accommodate recordings of a wide range of sampling frequencies, file durations and hydrophone configurations. In the shortlisted documents that used PAMGuard, data with a range of input sampling rates were processed (12-576 kHz), most of which were continuous, and some of which were duty cycled. PAMGuard offers built-in functionality to decimate audio (and can handle decimation to frequencies which are not an exact submultiple of the original frequency) and/or apply highpass, lowpass or bandpass filters. Click detector parameters (e.g., detection threshold, click duration) can be modified as desired. Hydrophone configurations can be explicitly specified and accounted for.

Instead of or in addition to reviewing detections in PAMGuard Viewer (see Manual Effort section below), exported detection files can be processed in external software, if desired. For example the R package PAMpal can be used for reading and processing PAMGuard data in R (Sakai et al., 2026).

Computation time

Based on one test dataset (Fundian Channel), computation time in PAMGuard was notably increased (59%) by enabling the click detector's echo detection. All further trials in PAMGuard were completed without echo detection. Processing speed varied across datasets, from 12-34x quicker than real time (Figure 2). Unsurprisingly, this is generally faster than reported in the literature for application of PAMGuard's click detector in combination with a MHT classifier; 10x real time for a sperm whale specific application (Fonollosa et al., 2024) and 4.8x real time for a multispecies application (Webber et al., 2022). Webber et al. (2022) highlighted that removing as many false positive clicks as possible before running the MHT was essential for efficiency of this

computationally intensive step. For the reasons discussed in the usability section, I did not assess computation times for LTSAs in PAMGuard. PAMGuard enables multi-threading by default.

Manual effort

Typically, PAMGuard outputs are reviewed in PAMGuard Viewer, to manually verify species presence. PAMGuard Viewer allows for listening to recordings, manually classifying detected clicks, and visualizing detected clicks in several ways.

Performance metrics

PAMGuard has been reported to achieve high positive detection rates compared to other software (Gubnitky and Diamant, 2024), but at the cost of high false positive rates (e.g., up to 86% in Webber et al., 2022). Use of PAMGuard's built in classifiers, such as the MHT classifier can help improve performance.

Software dependencies: Java-based

Ease of access: Freely available at pamguard.org

Additional consideration: PAMGuard's flexible detectors (including click, whistle and moan detectors) and classifiers allow it to be used to detect and classify a variety of cetacean species in a single processing run (Barlow et al., 2021; Young et al., 2025). It can also be used to localize species from multi-element arrays (and in doing so, estimate the number of individuals clicking). Plug-ins also expand functionality to allow additional analysis, such as estimation of inter-pulse intervals (IPIs) which allow body size estimation, allowing age-sex class classification (Miller et al., 2013).

Triton

Triton's most popular feature for detection of sperm whale clicks is the generation of LTSAs, though the software also offers a click detector, referred to as a 'SPICE detector'. The software is offered in uncompiled and compiled versions, which do and don't require a MATLAB license, respectively.

Useability

Triton is operated through a user-friendly GUI and has an informative user manual. User-developed software routines can be added to Triton (called 'Remoras') and while they are not detailed in the user manual they are described at github.com/MarineBioAcousticsRC/Triton/wiki. This includes the SPICE detector. However, when running the SPICE detector in the compiled version of Triton, no progress information or error messages are displayed, making it challenging to distinguish between when the software is still processing and when it has encountered an error. Thus, for use of the SPICE detector, good usability may require using the uncompiled version. Once during my trials (using the compiled version), I closed the software because it was being non-responsive, after which the software would not initiate at all without corrective intervention (see Appendix 1 for details). This resulted in a multi-hour trouble-shooting delay, but the issue was eventually resolved.

Flexibility

Triton is flexible to accommodate recordings of a wide range of sampling frequencies and file durations. In the shortlisted documents that used Triton, data with a range of input sampling rates

were processed (80-320 kHz), which were a mix of continuous and duty cycled recordings. Triton can decimate audio by whole-number factors. Click detector parameters (e.g., SNR, peak frequency limits, click duration) can be modified as desired. However, for an unknown reason, the SPICE detector only successfully processed one of my four sample datasets (Flemish Pass); for the other three data sets, the software reported "Problem reading file" and produced no further output.

Computation time

In a trial decimating one dataset (Fundian Channel) by 5x, LTSAs were generated 21% faster, but this was 15% slower overall, once the time taken to decimate the recordings was factored in. Decimating by 5x reduced the LTSA file size by 5x. Adjusting the temporal resolution of the LTSAs for this dataset from 5 s to 20 s also did not improve processing times (2.7% slower with 20 s resolution) but did reduce the LSTA files size by a factor of 3.9. All remaining trials were done with full bandwidth data and a 5 s resolution and resulting in processing speeds of 53-59x faster than real time.

For the one dataset that the SPICE detector successfully processed (Flemish Pass), processing time was significantly improved by decimating files ahead of time (67% time reduction, after factoring in time cost of decimation in Triton). Processing speed for this decimated dataset (including the time spent on decimation) was 25x faster than real time.

Manual effort

LTSAs require manual review to detect sperm whale presence. Typically, in shortlisted documents, that was done by examining LTSA windows that were 1 h long and displayed a frequency range between 0-30 kHz (for sperm whale specific analyses) and 0-100 kHz (for multispecies analyses) and then visually and aurally examining 5-10 s clips of audio to confirm the detection. This process is fairly streamlined in the Triton software interface but still requires non-negligible manual validation.

The SPICE detector produces TPWS (start Time, Peak-to-peak amplitude, Waveform, and Spectra parameters) files that can be used for manual annotation and classification in programs such as DetEdit. Additional Triton Remoras, such as the ClusterTool can aid in manual processing by using an unsupervised machine learning algorithm to automatically cluster similar clicks, which the analyst can classify as target species based on spectrogram features and ICIs (e.g., Constaratas et al., 2025).

Performance metrics

No performance metrics were reported in the shortlisted documents, comparing manual detections of sperm whale clicks in LSTAs relative to visual and aural review of whole datasets. While this process is likely to produce few false positives due to the manual review intrinsic to the process, it seems plausible that false negatives may occur, especially when clicks are faint.

Software dependencies: MATLAB Runtime R2018b (freely available)

Ease of access: Freely available at cetus.ucsd.edu/technologies_triton.html[2]

Additional considerations

Detection of sperm whales in Triton using LTSA and/or the SPICE detector can be incorporated into multispecies analyses.

DetEdit

DetEdit is a tool to visualize and annotate detected impulsive signals in large acoustic datasets, with capability to also detect those signals.

Useability

DetEdit can work with detections (in TPWS format) and LTSAs generated by Triton or can create its own. DetEdit has informative documentation available at github.com/MarineBioAcousticsRC/DetEdit, including step-by-step how-to guides. However, the process of getting everything formatted as required can still be challenging (A. Westell, personal communication, March 23, 2026). The DetEdit code I downloaded (March 9, 2026) did not execute successfully on the Triton output files I had generated; it produced an error that was beyond the scope of this report to troubleshoot. This error may have been due to a bug in the code, or an issue (e.g., formatting mismatch) in the dataset; DetEdit was designed to work with a 2014 version of Triton and a 2016 version of MATLAB (A. Westell, personal communication, Jan. 15, 2026), so some compatibility issues may be encountered if not using these software versions.

Solsona-Berga et al. (2020) reported processing a total of 202 TB of data (with 34 million sperm whale clicks), which provides evidence of capability to process large data volumes. However, in a very high data volume use case (40 continuous year-round recorders), it has been reported to not work stably and efficiently enough to keep up with processing needs (A. Westell, personal communication, Mar. 22, 2026).

Flexibility

In the shortlisted documents that used DetEdit, continuous data, with a sampling rate of 200 kHz was used. DetEdit should be flexible to accommodate recordings of a wide range of sampling frequencies and file durations.

Computation time

Since DetEdit is largely aimed at providing tools for manual review and validation, and can leverage Triton outputs, I did not conduct processing time trials for DetEdit.

Manual effort

One of the key abilities of DetEdit is enabling efficient manual validation and classification of detections. Subjectively, validation is reported to be fast in DetEdit (A. Westell, personal communication, Jan. 15, 2026).

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[2] This compiled, standalone version of Triton may not have all of the capabilities of the full Triton code base (which requires MATLAB licence). Source code can be found at github.com/MarineBioAcousticsRC/Triton

Performance metrics

No performance metrics were reported in the shortlisted documents, but DetEdit can be used to determine detectors' false positive rates.

Software dependencies: MATLAB (standard pricing is $1050/license/year)

Ease of access: Freely available from github.com/MarineBioAcousticsRC/DetEdit

CABLE

CABLE is focused not only on detection of sperm whale clicks but also identifying "good" clicks that it uses to calculate accurate IPI distributions. These IPI distributions are then used to estimate how many sperm whales are present in a recording, and how large they are.

Useability

CABLE is operated through a user friendly GUI, has a detailed user manual and is reasonably straightforward to understand and implement. No technical issues were encountered during trial runs.

Flexibility

CABLE's click classifier was trained on data with a frequency range of 2-12kHz, and automatically filters and resamples recordings to fit its requirements. So, it can process data from a range of sampling rates, but may perform poorly with recordings lacking the full extent of this frequency range. The extent to which this is a problem or not has not been evaluated.

Computation time

CABLE's computation speed varied substantially with the number of clicks it detected (Pearson correlation coefficient: 0.97); it sometimes outperformed most other methods but was the slowest of all methods when processing data with many clicks detected (Figure 2). For example, though the Fundian Channel and Gully datasets are of identical sampling frequencies (Table A1) and lengths (Table A2), processing the Fundian Channel dataset, which had 44x as many clicks detected by CABLE, was 5x slower (Figure 1). This means processing times can be hard to predict.

Manual effort

No manual intervention is required for use of CABLE. However, users wanting to strictly confirm sperm whale presence could use CABLE outputs (.csv files with timestamps of "good" clicks) to find times for manual review in external software. The number of "good" clicks found in a file could be used to check the most promising files in a day first, reducing manual effort required.

Performance metrics

The only published performance metrics for CABLE assessed the classification of on-axis (aka "good") sperm whale clicks, rather than sperm whale clicks in general (Beslin et al., 2018). For this assessment, true positive rate was 89.1%, true negative rate was 99.5%, and precision was 92.7%. However, the proportion of detected clicks accepted as "good" is always very low (often < 1%). Thus, it is possible that sperm whale clicks may be detected but all rejected, especially for short

duration recordings. However, I would expect that for analyses interested in presence at a daily resolution, unless SNRs are low and the encounter brief, this is unlikely to result in false negatives at this scale, due to the high rate at which clicks are produced by sperm whales.

Software dependencies: MATLAB Runtime R2015a (freely available)

Ease of access: Freely available at whitelab.biology.dal.ca/CABLE/cable.htm

Additional considerations

CABLE provides built-in body size estimation, allowing age-sex class estimation. However, CABLE's stringent criteria for good clicks may apply more strictly (potentially leading to false negatives) on large males, since few were included in the training data. However, the extent to which this is a problem or not has not been evaluated, and CABLE has been used to detect adult males off of Ireland (Barile et al., 2024).

Raven Pro

Raven Pro's BLED detects events where the energy in a specified frequency band exceeds a threshold for a specified range of durations

Useability

Raven Pro is operated through a user-friendly GUI, has video tutorials and other documentation online, including multiple resources related to use of BLEDs. Some lag was the only technical issue encountered during trial runs.

Flexibility

Raven Pro is not directly restricted in the range of sampling frequencies and file durations it can process. However, I noticed a lag when playing recording with a sampling frequency of 256 kHz. This delay could be alleviated by decimating the recordings in external software (e.g., Triton) prior to viewing in Raven Pro.

Computation time

Based on one test dataset (Fundian Channel), computation time in Raven Pro was significantly improved by decimating files ahead of time (56% time reduction, after factoring in time cost of decimation in Triton). So, all further trials in Raven Pro were completed using the decimated files. For compatibility, computation times reported include the time spent on decimation. Even so, Raven Pro's BLED consistently had the fastest processing times, at 56-88x real time. However, detector settings were not necessarily optimized for these datasets, and I did not test whether processing times varied with detector settings.

Manual effort

Raven Pro detections can be manually reviewed within the Raven Pro interface. Raven Pro documentation emphasizes that getting good results using a BLED is an iterative process. Oestreich et al. (2024) also mentioned manual tuning of parameters to optimize detection. Also see relevant comments in Performance Metrics section below.

Performance metrics

Raven Pro's BLED is non-specific in that it ignores the shape and pattern of a sound. As such, it can have a high false positive rate, which Oestreich et al. (2024) addressed by applying an ICI based filter after processing in Raven Pro. After filtering for ICIs of 0.5-2 s, with 6 repetitions (rounded to the nearest 0.25 s), Oestreich et al. (2024) achieved precision of 96%, recall of 96%, and a false positive rate of 4% (daily resolution). Achieving optimal performance requires comparison to manually identified clicks to select the repetition threshold, as this significantly affected performance metrics.

Software dependencies: Java-based

Ease of access

Raven Pro is paid software ($200 USD/license/year), which can be downloaded at ravensoundsoftware.com. However, Raven Workbench is a collection of software modules that will gradually replace and augment Raven Pro's functionality, which is in the works and will be distributed free of charge. Once Raven Workbench has replaced Raven Pro, Raven Pro will be available free of charge (but with no future updates or user support).

Gubnitky and Diamant (2024) custom MATLAB script

This method is specifically designed for detection of sperm whale echolocation clicks in noisy environments that may include clicks from multiple sperm whales and other noise inputs.

Useability

While the source code is available, there is essentially no publicly available documentation and supporting materials. The code I downloaded (March 9, 2026) did not execute successfully using default settings on a small set of test files from the Flemish Pass dataset. The error may be related to file length not being exact multiples of the buffer length used to look for clicks, but this was beyond the scope of this report to troubleshoot in full.

Flexibility

This method uses a 2-24 kHz bandpass filter; Gubnitky and Diamant (2024) do not directly comment on the method's generalizability to recordings with more limited frequency bandwidths.

Computation time

Not assessed (see Useability).

Manual effort

No manual review is described.

Performance metrics

In trials comparing this method and PAMGuard's click detector, Gubnitky and Diamant (2024) reporter high precision for this method, even without pre-training for new datasets.

Software dependencies

MATLAB (standard pricing is $1050/license/year), and MATLAB's Signal Processing Toolbox and Statistics and Machine Learning Toolbox ($526 and $550 USD/ license/year, respectively).

Ease of access

Freely available at github.com/Project-CETI/Sperm_whale_click_presence_detector

Bermant et al. (2019) custom Python script

Overall, I did not get the impression that this convolutional neural network-based detector for sperm whale clicks is being put forward as a production ready option. For example, Bermant *et al.* (2019) point out possible misclassifications in their training dataset, to be addressed in a more comprehensive study. Thus, I include this method as a proof of concept of an approach that may be of interest in the future, as user-ready tools are built.

Useability

While the source code is available, there is essentially no publicly available documentation and supporting materials.

Flexibility

This method uses a 2-20 kHz bandpass filter; Bermant *et al.* (2019) do not directly comment on the method's generalizability to recordings with more limited frequency bandwidths.

Computation time

Not assessed, because I did not believe the method to be production ready.

Manual effort

No manual review is described.

Performance metrics

Bermant *et al.* (2019) report 100% detector success on a limited dataset, but did not specify whether other signals (e.g., vessel noise or other species) were included.

Software dependencies: Python 3 (freely available)

Ease of access: Freely available at github.com/dgruber212/Sperm_Whale_Machine_Learning

Recommendations and Conclusions

All methods considered in this report have strengths and weaknesses and there is no one best software for sperm whale click detection and classification across all scenarios. Considering the click detectors assessed, all approaches face trade-offs between rates of false negatives and false positives, and improving accuracy often comes with increased processing time. Using LTSAs also has trade-offs; while generating LTSAs in Triton was relatively quick, post-processing requires manual review of all outputs. Below, I discuss strengths, weaknesses, implementation advice and other considerations for the four methods that I successfully tested across all four datasets.

LTSAs in Triton may be an effective choice, despite mandatory manual review, especially for research groups that are already using this tool for other purposes. Given that Triton is currently used in DFO Maritimes Region workflows for detection of other species, adding sperm whale detection to existing workflows may be an effective choice. There may be opportunities to leverage existing software familiarity and processing runs to detect sperm whales. Triton's LTSAs show promise for sperm whale detection, given their established use for this task in the literature, predictable and efficient processing speed (Figure 2), free of charge implementation, and user-friendly interface. Extensive manual review is one possible downside of this approach. Using LTSAs in combination with a click detector may help refine the amount of data to be reviewed (e.g., via DetEdit), but this trades off with additional computation requirements. Another possible way to reduce manual effort (at the cost of increased computation time) could be to train a machine learning-based detector on LTSAs, to detect periods with likely sperm whale clicks.

Raven Pro's BLED also stands out for its fast processing speed (Figure 2) and good usability. However, it is a less established tool for detecting sperm whale clicks and currently requires a paid license for use, albeit not as expensive as MATLAB, and with a free version likely available in coming years. It remains untested to what extent processing speed varies with detector parameters, and thus it is uncertain what speeds will be realized once the detector is tuned to the input datasets. Thus, this method shows promise, but I would advise further testing of performance and of sensitivity of processing times to detector parameters before pursuing full scale implementation.

While PAMGuard was relatively slow and its interface is complex, it is a free, well-established, powerful, and extendable tool for sperm whale click detection. With flexible click detectors, click train detectors, classifiers, and plug-ins that expand functionality (e.g., IPI calculation), PAMGuard is versatile. It allows research groups room to grow if they start out with a simple objective (e.g., detect presence/absence on a daily timescale), but want to take their analysis further in the future. Additionally, PAMGuard has the capacity to concurrently detect and classify multiple species, which could be of interest for research programs with a multi-species scope. Since PAMGuard's click detector often has high rates of false positives, subsequent classification is likely needed for efficient manual validation. This could be done through PAMGuard's built in classifiers (as in Fonollosa et al., 2024 and Webber et al., 2022), or external tools. For example, SoundScope is a python-based software for visualization and manual verification of detections, currently being developed at the Passive Acoustic Branch of the National Oceanic and Atmospheric Administration's Northeast Fisheries Science Center, with the intent to use it in conjunction with PAMGuard (Mouy, 2024/2026).

As a free, easy-to-implement tool with widely variable processing speeds, CABLE also has its strengths and weaknesses. Due to its stringent "good click" criteria, CABLE should have lower false positive rates than other methods considered, potentially requiring less manual review. However, the evaluation it does to achieve this results in slow processing speeds when many clicks are detected. One approach to mitigate this issue in an analysis of daily presence could be to analyse the first hour of each day and only more on to processing each subsequent hours if no sperm whales were detected in the previous hour. Also, whether in some cases all detected clicks are rejected, resulting in false negatives, and the extent to which this potential issue could be offset by relaxing parameter strictness, requires evaluation. It should be noted, however, that rejected click detections are reported in output logs at a per-file resolution, which would allow these rejected

clicks to still be used to screen for recordings with potential sperm whale presence in the absence of “good” clicks. Lastly, while none of the methods this section (including CABLE) has formal constraints on the sampling frequencies of recordings they will process, CABLE may not perform as expected when frequency bandwidths do not include the 2-12 kHz range that that software was trained on.

While reviewing literature for this report, it was apparent that there is active work to develop new methods and improve existing ones for detection and classification of cetaceans, including sperm whales specifically. As such, new approaches, or tools to enhance existing approaches, will be worth monitoring.

Limitations

The recommendations of this report are based on several constraints as described in the assessment approach. One screening criteria that filtered out many approaches was focusing my detailed consideration on methods with a readily available implementation. This approach was taken as I had in mind teams/researchers who do not have time and/or expertise to invest in development of implementations of described approaches and prefer proven and readily available options. For teams/researchers who are not facing this constraint, a more in-depth review may be warranted of documents screened out based on this criterion.

Additionally, relative processing speeds may not generalize across different computers and workflows. The desktop computer used for the trials conducted here was relatively well equipped (CPU: Processor AMD Ryzen 9 7950X3D 16-Core Processor, Memory: 96 GB 6000 MT/s DDR5, Disk: Samsung SSD 990 EVO Plus 1TB, GPU: AMD Radeon RX 9070 [16GB GDDR6 memory]). Also, while some software (e.g., CABLE) use multithreading by default, I did not test parallel processing of multiple concurrent software instances, which may allow faster data processing overall in some instances.

Extensions and Future Applications

Effective detection of sperm whale clicks in large PAM datasets will open the door to further analyse these datasets and extend the utility of PAM for sperm whales. This could include identifying cultural groups using vocalizations called codas (Rendell and Whitehead, 2003), by running coda detectors (e.g., Gubnitsky et al., 2025) on periods for which sperm whale presence has been detected. This is particularly valuable given the importance of culture to effective sperm whale conservation (Eguiguren et al., 2025). Further, using IPIs to approximate body size and, by extension, the age/sex class of individual sperm whales (Beslin et al., 2018; Miller et al., 2013) can provide a wealth of information for population monitoring.

Data and Code Availability

For further details, including full lists of publications reviewed, per-document screening outcomes and code for calculating summary statistics, see:
github.com/GlobalSpermWhaleProject/AcousticDetectionMethodEvaluation

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References

Barile, C., Berrow, S., & O'Brien, J. (2024). Click-click, who's there? Acoustically derived estimates of sperm whale size distribution off western Ireland. *Frontiers in Marine Science*, *10*. https://doi.org/10.3389/fmars.2023.1264783

Barlow, J., Cheeseman, T., & Trickey, J. S. (2021). Acoustic detections of beaked whales, narrow-band high-frequency pulses and other odontocete cetaceans in the Southern Ocean using an autonomous towed hydrophone recorder. *Deep Sea Research Part II: Topical Studies in Oceanography*, *193*, 104973. https://doi.org/10.1016/j.dsr2.2021.104973

Bermant, P. C., Bronstein, M. M., Wood, R. J., Gero, S., & Gruber, D. F. (2019). Deep machine learning techniques for the detection and classification of sperm whale bioacoustics. *Scientific Reports*, *9*, 12588. https://doi.org/10.1038/s41598-019-48909-4

Beslin, W. A. M. (2018). *Measuring Inter-Pulse Intervals in Sperm Whale Clicks: Development of an Automatic Method and Its Potential for Identifying Eastern Caribbean Social Units* (Master of Science; pp.149pp). Dalhousie University, Halifax, Nova Scotia.

Beslin, W. A. M., Whitehead, H., & Gero, S. (2018). Automatic acoustic estimation of sperm whale size distributions achieved through machine recognition of on-axis clicks. *The Journal of the Acoustical Society of America*, *144*(6), 3485–3495. https://doi.org/10.1121/1.5082291

Constaratas, A. N., Holcer, D., Özgöbek, Ö., & Širović, A. (2025). Acoustic occurrence of deep-diving cetaceans in the southern Adriatic Sea. *Marine Mammal Science*, *41*(2), e13204. https://doi.org/10.1111/mms.13204

Eguiguren, A., Avila, I., Mesnick, S., Cantor, M., Hersh, T. A., Pe/rez-Puig, H., Rosero, P., Rendell, L., Whitehead, H., Rojas, C., & Alava, J. J. (2025). Integrating cultural dimensions in sperm whale (*Physeter macrocephalus*) conservation: threats, challenges, and solutions. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *380*, 20240142. https://doi.org/10.1098/rstb.2024.0142

Feyrer, L. J., Colbourne, N., Lawson, J. W., Moors-Murphy, H. B., & Ferguson, S. H. (2025). Three decades of observer records reveal ongoing risks of marine mammal depredation and entanglement in Canada's Atlantic fisheries. *ICES Journal of Marine Science*, *82*(8), fsaf140. https://doi.org/10.1093/icesjms/fsaf140

Fonollosa, L. G., Webber, T., Brotons, J. M., Cerda, M., Gillespie, D., Pirotta, E., & Rendell, L. (2024). Comparing neural networks against click train detectors to reveal temporal trends in passive acoustic sperm whale detections. *Journal of the Acoustical Society of America*, *156*(6), 4073–4084. https://doi.org/10.1121/10.0034602

Gillespie, D., Macaulay, J., Oswald, M., & Roch, M. (2026). PAMGuard: Application software for passive acoustic detection, classification, and localisation of animal sounds. *The Journal of the Acoustical Society of America*, *159*(1), 437–443. https://doi.org/10.1121/10.0042245

Gomez, C., Konrad, C. M., Vanderlaan, A., Moors-Murphy, H. B., Marotte, E., Lawson, J. W., Kouwenberg, A.-L., Fuentes-Yaco, C., & Buren, A. (2020). *Identifying priority areas to enhance monitoring of cetaceans in the Northwest Atlantic Ocean*. Dartmouth, Nova Scotia: Fisheries and Oceans Canada.

Gracic, M., Gubnitsky, G., & Diamant, R. (2025). A survey of detection techniques for sperm whale and dolphin echolocation clicks. *Frontiers in Marine Science*, *12*. https://doi.org/10.3389/fmars.2025.1567001

Gubnitky, G., & Diamant, R. (2024). Detecting the presence of sperm whales' echolocation clicks in noisy environments. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *32*, 2050–2061. https://doi.org/10.1109/TASLP.2024.3379899

Gubnitsky, G., Mevorach, Y., Gero, S., Gruber, D. F., & Diamant, R. (2025). Automatic detection and annotation of eastern Caribbean sperm whale codas. *Scientific Reports*, *15*(1), 12790. https://doi.org/10.1038/s41598-025-97009-z

K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology. (2026). *Raven Pro: Interactive Sound Analysis Software*. Retrieved from https://www.ravensoundsoftware.com/.

Madsen, P., Wahlberg, M., & Møhl, B. (2002). Male sperm whale (*Physeter macrocephalus*) acoustics in a high-latitude habitat: implications for echolocation and communication. *Behavioral Ecology and Sociobiology*, *53*(1), 31–41. https://doi.org/10.1007/s00265-002-0548-1

Mellinger, D. K., Stafford, K. M., & Fox, C. G. (2004). Seasonal occurrence of sperm whale (*Physeter macrocephalus*) sounds in the Gulf of Alaska, 1999–2001. *Marine Mammal Science*, *20*(1), 48–62. https://doi.org/10.1111/j.1748-7692.2004.tb01140.x

Miller, B. S., Growcott, A., Slooten, E., & Dawson, S. M. (2013). Acoustically derived growth rates of sperm whales (*Physeter macrocephalus*) in Kaikoura, New Zealand. *Journal of the Acoustical Society of America*, *134*(3), 2438–2445. https://doi.org/10.1121/1.4816564

Mouy, X. (2026). *SoundScope* [Python]. Retrieved from https://github.com/xaviermouy/SoundScope (Original work published 2024)

Oestreich, W. K., Benoit-Bird, K. J., Abrahms, B., Margolina, T., Joseph, J. E., Zhang, Y., Rueda, C. A., & Ryan, J. P. (2024). Evidence for seasonal migration by a cryptic top predator of the deep sea. *Movement Ecology*, *12*(1), 65. https://doi.org/10.1186/s40462-024-00500-x

Rendell, L. E., & Whitehead, H. (2003). Vocal clans in sperm whales (*Physeter macrocephalus*). *Proceedings of the Royal Society of London. Series B: Biological Sciences*, *270*(1512), 225–231. https://doi.org/10.1098/rspb.2002.2239

Sakai, T., Barlow, J., Griffiths, E., Oswald, M., Baumann-Pickering, S., & Oswald, J. (2026). *PAMpal: Load and Process Passive Acoustic Data*. Retrieved from https://cran.r-project.org/web/packages/PAMpal/index.html

Solsona-Berga, A., Frasier, K. E., Baumann-Pickering, S., Wiggins, S. M., & Hildebrand, J. A. (2020). DetEdit: A graphical user interface for annotating and editing events detected in long-term acoustic monitoring data. *PLOS Computational Biology*, *16*(1), e1007598. https://doi.org/10.1371/journal.pcbi.1007598

Stanistreet, J. E., Nowacek, D. P., Bell, J. T., Cholewiak, D. M., Hildebrand, J. A., Hodge, L. E. W., Parijs, S. M. V., & Read, A. J. (2018). Spatial and seasonal patterns in acoustic detections of sperm whales *Physeter macrocephalus* along the continental slope in the western North Atlantic Ocean. *Endangered Species Research*, *35*, 1–13. https://doi.org/10.3354/esr00867

Taylor, B. L., Baird, R., Barlow, J., Dawson, S. M., Ford, J., Mead, J. G., Notarbartolo di Sara, G., Wade, P., & Pitman, R. L. (2019). *Physeter macrocephalus* (amended version of 2008 assessment). *The IUCN Red List of Threatened Species 2019*, *e.T41755A160983555*. https://doi.org/10.2305/IUCN.UK.2008.RLTS.T41755A160983555.en

Webber, T., Gillespie, D., Lewis, T., Gordon, J., Ruchirabha, T., & Thompson, K. F. (2022). Streamlining analysis methods for large acoustic surveys using automatic detectors with operator validation. *Methods in Ecology and Evolution*, *13*(8), 1765–1777. https://doi.org/10.1111/2041-210X.13907

Whitehead, H., & Shin, M. (2022). Current global population size, post-whaling trend and historical trajectory of sperm whales. *Scientific Reports*, *12*(1). https://doi.org/10.1038/s41598-022-24107-7

Young, K. F., Webber, T., Karantzas, L., Miteva, S., Oakes, G., Santillo, D., & Johnston, P. (2025). Threatened cetaceans in a potential deep seabed mining region, Clarion Clipperton Zone, Eastern Pacific, August 2023. *Frontiers in Marine Science*, *12*. https://doi.org/10.3389/fmars.2025.1511075

Appendix 1

Table A1: Sample of acoustic datasets, in Canadian waters, owned by DFO or by academic research groups collaborating with Christine Clarke. This is not an exhaustive list, but generally captures the range of parameters in datasets of relevance for detection of sperm whale. Datasets used to test shortlisted methods for detection and classification of sperm whale clicks are highlighted in blue (see Table A2).

Deployment Date	Location[3]	Owner	Type	Recorder[4]	Sampling Freq. (kHz)	Duty Cycle (s)	Deployment Length (days)
2023-08-25	Fundian Channel 41.7691˚N, 65.575˚W	DFO	Bottom-mounted	JASCO AMAR G4	256	370 / 900	377
2022-10-10	The Gully 43.718˚N, 58.7221˚W	DFO	Bottom-mounted	JASCO AMAR G4	256	370 / 900	316
2014-07-14	Logan Canyon 43.451˚N, 59.950˚W	DFO	Bottom-mounted	JASCO AMAR G3	16 250	680 / 900 195 / 900	76
2013-05-08	Shortland-Haldimand 44.098˚N, 58.056˚W	DFO	Bottom-mounted	JASCO AMAR G3	32 128	780 / 900 120 / 900	141
2025-07-16	Bonavista Bay 48.611˚N, 53.479˚W	DFO	Bottom-mounted	SoundTrap (ST400)	128	Continuous	26
2017-06-18	Explorer Seamount 48.994˚N, 130.901˚W	DFO	Bottom-mounted	JASCO AMAR G3	16 250	341 / 900 60 / 900	384
2024-05-31	Offshore Gowgaia Bay 52.394˚N, 131.712˚W	DFO	Bottom-mounted	JASCO AMAR G4	48	300 / 900	~ 365
2024-06-03	Offshore Cape Knox 54.249˚N, 134.024˚W	DFO	Bottom-mounted	JASCO AMAR G4	24	300 / 900	~ 365
2025-08-12	Scotian Shelf Edge 58.318˚N 44.018˚W	Whitehead Lab	Vessel-towed	Custom two-element array	192	Continuous	9
2025-06-25	Flemish Pass 47.809˚N 48.959˚W	Whitehead Lab	Vessel-towed	Custom two-element array	192	Continuous	15
2023-08-30	Baffin Bay 59.010˚N 67.664˚W	Fortune Whale Lab	Glider	DMON2	96	60 / 300	45
2022-09-12	Baffin Bay 59.811˚N 68.553˚W	Fortune Whale Lab	Glider	DMON1	60	60 / 180	32

[3] For vessel-towed or glider data, the location listed is for the mid-point of the track for the dataset.
[4] All single channel, except two-channel 'Custom two-element array'.

Table A2: Sample datasets used for testing shortlisted methods for detection and classification of sperm whale clicks. See Table A1 for full deployment details.

Location	Dates	Days	Per File Length	Per File Size (mb)	Number of Files	Size of Dataset Sample (GB)	Length of Dataset Sample (hrs)
Fundian Channel	Sep. 8-14, 2023	7	6m07s	276	672	176	69
The Gully	May 1-7, 2023	7	6m09s	277	672	177	69
Bonavista Bay	Jul. 21-27, 2025	7	5m00s	75	2016	144	168
Flemish Pass	Jun. 28-30, 2025	3	9m42s; 17s[5]	655; 19	439	139	37[6]

[5] This dataset is generally made up of alternating 9m42s and 17s files (both at the same sampling frequency)

[6] Available data for this period was less than the expected 72 hrs, due to gaps in the audio data.

Table A3. Software versions used and parameter settings for trial acoustic detection of sperm whales. These settings were selected as generally appropriate based on use in the shortlisted documents and other literature on sperm whale click parameters, but they were not specifically optimized for the datasets in use.

Software	Version	Detection Method	Parameters	
PAMGuard	2.02.18	Click detector	See Figure A2. Default click detector parameters used except where otherwise specified. Click length parameters[7] Min Click Separation: 960 samples Pre sample: 480 samples High pass filtered (4th order Butterworth 1 kHz) Decimated to 48 kHz (default low pass filter: 24 kHz; using linear interpolation[8])	 Max Click Length: 1920 samples Post sample: 480 samples
Triton	1.93.20160524 (compiled)	LTSA	5s 100 Hz resolution	
	1.93.20160524 (uncompiled[9])	SPICE detector	SNR threshold: 10 dB Duration range: 25-1200 μs Remove isolated clicks: >10 s	Bandpass filter: 0.5-24 kHz Peak energy range: 1-24 kHz Remove echoes: <0.001 s
CABLE	1.0		Main Menu setting[10]: nRep = 1 Advanced settings[11]: IPIRange: 2 12 IPItol: 0.1	Goodness = 0.1 ICIRange: 0.20 10 maxIPIDeviation: 0.1
Raven Pro	1.6	BLED	As specified in Oestreich et al. (2024)	

[7] Based on a maximum click length of 40 ms, as used by Beslin (2018), and an assumed typical click length of 20 ms

[8] When applicable, i.e., for recordings that did not have a starting sampling rate that was a direct multiple of 48 kHz.

[9] Using MATLAB version R2025b.

[10] Selected in consultation with W. Beslin to balance excluding bad clicks, while being flexible on "goodness", since the software was not trained on much or any male data (which will differ in some parameters assessed, such as click duration) but recordings from the North Atlantic are expected to have substantial acoustic presence from adult males.

[11] Advanced settings were also adjusted in consultation with W. Beslin to allow the software to include clicks of males up to 21m length (max IPI), to permit less precise IPI measurements that will still enable age-sex class discrimination (maxIPIDeviation and IPItol), and to account for the software potentially excluding some clicks in a train due to limited training on male data (ICIRange). For all settings not specified, default values were used.

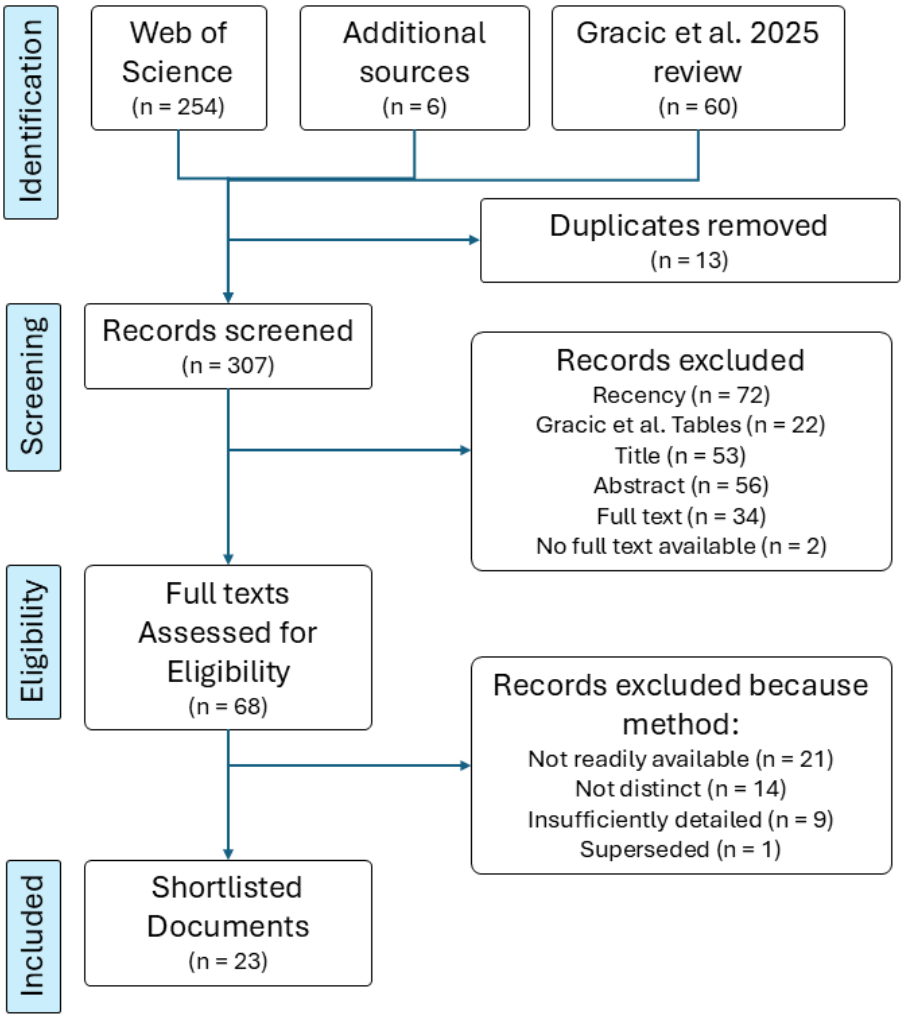

Figure A1: Diagram of process of literature review, screening, eligibility assessment and record inclusion, for assessment of methods of detection and classification of sperm whale clicks in passive acoustic datasets.

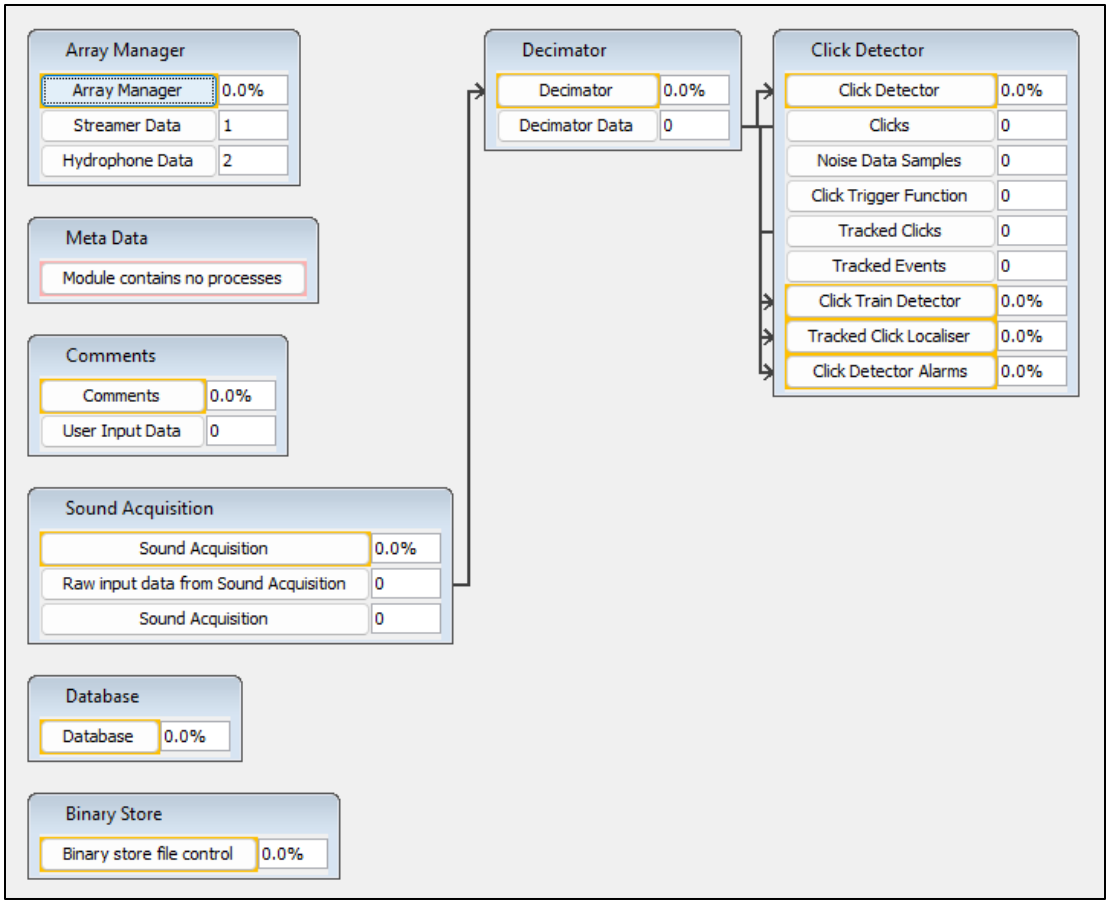

Figure A2: PAMGuard object model used for software trials.

Triton troubleshooting details

Once during my trials, I closed Triton (compiled version) because it was non-responsive, after which the software would not initiate at all; the splash screen would temporarily appear then disappear. Trouble shooting steps that did not resolve the issue included: restarting the computer, uninstalling and reinstalling Triton and MATLAB Runtime, clearing potentially relevant files from User Temp folders (though after the issue was resolved, additional cache files were found, at C:\Users\Christine\AppData\Local\Temp\ Christine\mcrCache9.5, that perhaps would have resolved the issue if deleted). No relevant error logs were found in Windows logs at Application or System level, and no diagnostic text or error message was displayed when initiating Triton from Command Prompt (which also failed to launch after displaying splash screen). The issue was eventually resolved by creating a new folder (C:\ProgramData\MathWorks\MCRCache) and adding a new System Variable (Name: MCR_CACHE_ROOT; Value: C:\ProgramData\MathWorks\MCRCache) to the computer's Environment Variables to force MATLAB Runtime cache to this location. Triton itself may be stable in normal use, but it appears that a hang followed by forced restart can leave the MATLAB Runtime cache/temp files in a corrupted or unusable state that survives reinstall of the app. Overall, this resulted in a multi-hour trouble-shooting delay, as the environment-level intervention required was not obvious. This kind of intervention may also be challenging to implement on devices where users have limited admin privileges.